

By **Law I** the body leaving B would run straight on to c (so Bc continues the line AB), yet it turns to C : some impulse deflected it by cC . Now $\triangle SBc = \triangle SAB$ because Sc and AB share the base SB and Bc lies on AB ; but we are **GIVEN** $\triangle SBC = \triangle SAB$, hence $\triangle SBC = \triangle SBc$. By **Euclid I.40**, equal triangles on the same base SB lie between the same parallels, so C and c are equidistant from line SB — that is, $cC \parallel BS$. The deflecting impulse (by **Law II**, along cC) therefore points exactly along BS , straight toward the centre.